

Komalpreet Kaur

AI/ML Engineer

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Projects

Vanta | PyTorch, SpeechBrain, FastAPI, Next.js, Docker, Hugging Face Mar 2026 - Apr 2026

- Built a Target Speaker Extraction system isolating a target voice from noisy recordings using a 5-second reference clip.
- Trained Conv-TasNet from scratch with SI-SDR loss; paired with frozen ECAPA-TDNN encoder for voice conditioning.
- Deployed FastAPI backend on Hugging Face Spaces and Next.js frontend on Vercel.
- Repository Link: <https://github.com/Komalpreet2809/Vanta>

Soma | FastAPI, React, LangChain, Neo4j, ChromaDB, SQLite, Docker Mar 2026 - Apr 2026

- Designed a brain-inspired AI system with multi-layered memory architecture for context-aware reasoning.
- Implemented hybrid RAG pipeline combining ChromaDB and Neo4j with LangGraph for multi-step AI decision-making.
- Built full-stack system with FastAPI and React for consolidation and knowledge refinement.
- Repository Link: <https://github.com/Komalpreet2809/SOMA>

Specula | Python, FastAPI, PyTorch, HuggingFace, OpenCV Feb 2026 - Mar 2026

- Developed a full-stack AI tool using a 5-method forensic pipeline for deepfake image detection.
- Implemented CNN classification with Grad-CAM explainability and analyzers like ELA, FFT, and metadata forensics.
- Designed a weighted scoring system to generate confidence-based authenticity verdicts and enabled batch + PDF reporting APIs.
- Repository Link: <https://github.com/Komalpreet2809/Specula>

Qlothi | Python, FastAPI, PyTorch, HuggingFace, OpenCV, JavaScript Jan 2026 - Mar 2026

- Built a Chrome extension for Pinterest enabling AI-based outfit segmentation and product discovery.
- Integrated SegFormer model for clothing detection and automated visual search using Playwright + Google Lens.
- Developed FastAPI backend and interactive UI with filtering features for style, budget, and category-based recommendations.
- Repository Link: <https://github.com/Komalpreet2809/Qlothi>

Certifications

Master Gen AI & Gen AI tools Jul 2025 - Aug 2025

Udemy — [Certificate Link](#)

Social Networks Jan 2025 - Apr 2025

NPTEL — [Certificate Link](#)

Skills

Languages: Python, Java, C++, Bash

Machine Learning & AI: Supervised & Unsupervised Learning, Deep Learning, Computer Vision, NLP, Feature Engineering

Generative AI & RAG: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, OpenCV, NumPy, Pandas

Backend & APIs: FastAPI, REST APIs, Model Deployment

Frontend & Tools: React, JavaScript, HTML, CSS, Chrome Extensions

Databases: ChromaDB (Vector DB), Neo4j (Graph DB), SQL (SQLite), NoSQL

MLOps & Systems: Docker, Git, GitHub Actions, CI/CD, Linux

Cloud & Platforms: AWS, Azure

Education

Lovely Professional University 2023 – 2027

Bachelor of Technology - Computer Science and Engineering — CGPA: 8.56

Phagwara, Punjab

Shri Rajendra Giri Memorial Academy 2021 – 2022

12th — Percentage: 93%

Kheri, Uttar Pradesh

Shri Guru Nanak Dev Public School 2019 – 2020

10th — Percentage: 96.2%

Kheri, Uttar Pradesh